

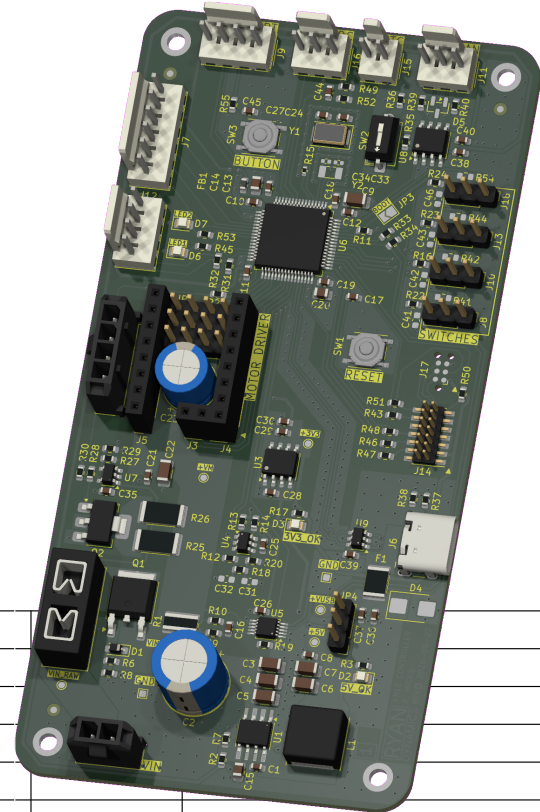
KNEE CONTROLLER

VARIANT: PRELIMINARY

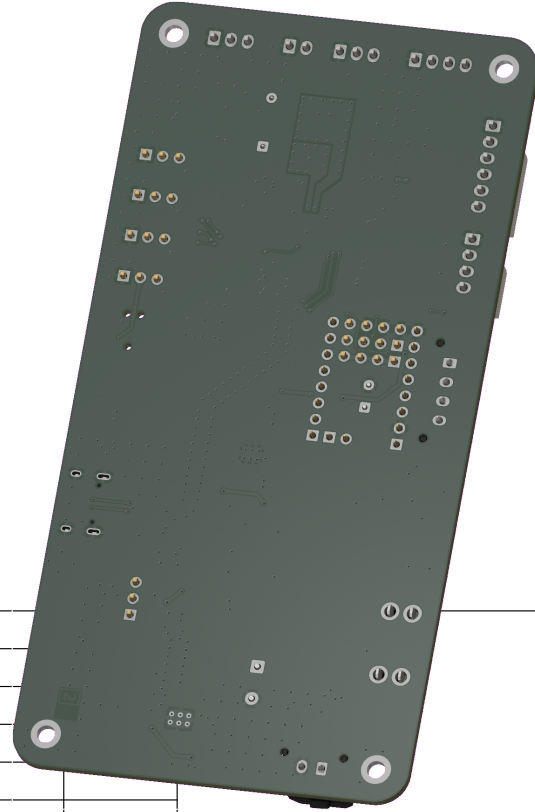
2026-10-09
REV A

Page	Index	Page	Index	Page	Index	Page	Index
1	COVER PAGE	11		21		31	
2	ASSEMBLY OVERVIEW	12		22		32	
3	TOP ASSEMBLY	13		23		33	
4	BOTTOM ASSEMBLY	14		24		34	
5	TOP FABRICATION	15		25		35	
6	BOTTOM FABRICATION	16		26		36	
7	L1-L2 DRILL MAP	17		27		37	
8	TOP TESTPOINTS	18		28		38	
9	L1 COPPER	19		29		39	
10	L2 COPPER	20		30		40	

TOP VIEW



BOTTOM VIEW



NOTES

COMMENT

NOT FITTED COMPONENTS MARKED AS **X**

DRAFT – EARLY STAGE SCHEMATIC CAPTURE

PRELIMINARY – LATE STAGE SCHEMATIC CAPTURE

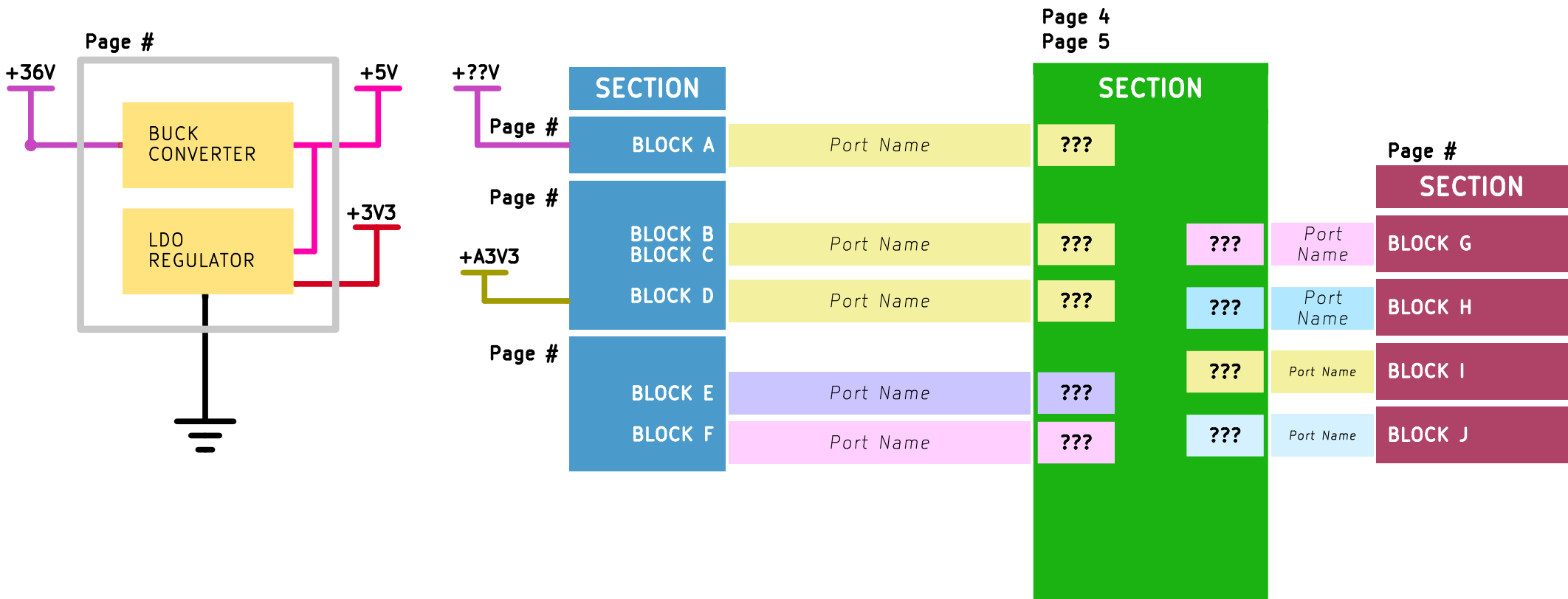
CHECKED – PCB LAYOUT CHECKED, CONTACT ENGINEER IF MISTAKE FOUND

RELEASED – READY FOR PRODUCTION

	DRAWN	CHECKED	APPROVED	BOARDWRIGHT	
NAME	R. HICKS				
DATE	2026-09-27				
SCALE NTS	SCHEMATIC			KNEE CONTROLLER	
				DRG. NO.	REV
	A3	SHEET 1 OF 10			4840.703

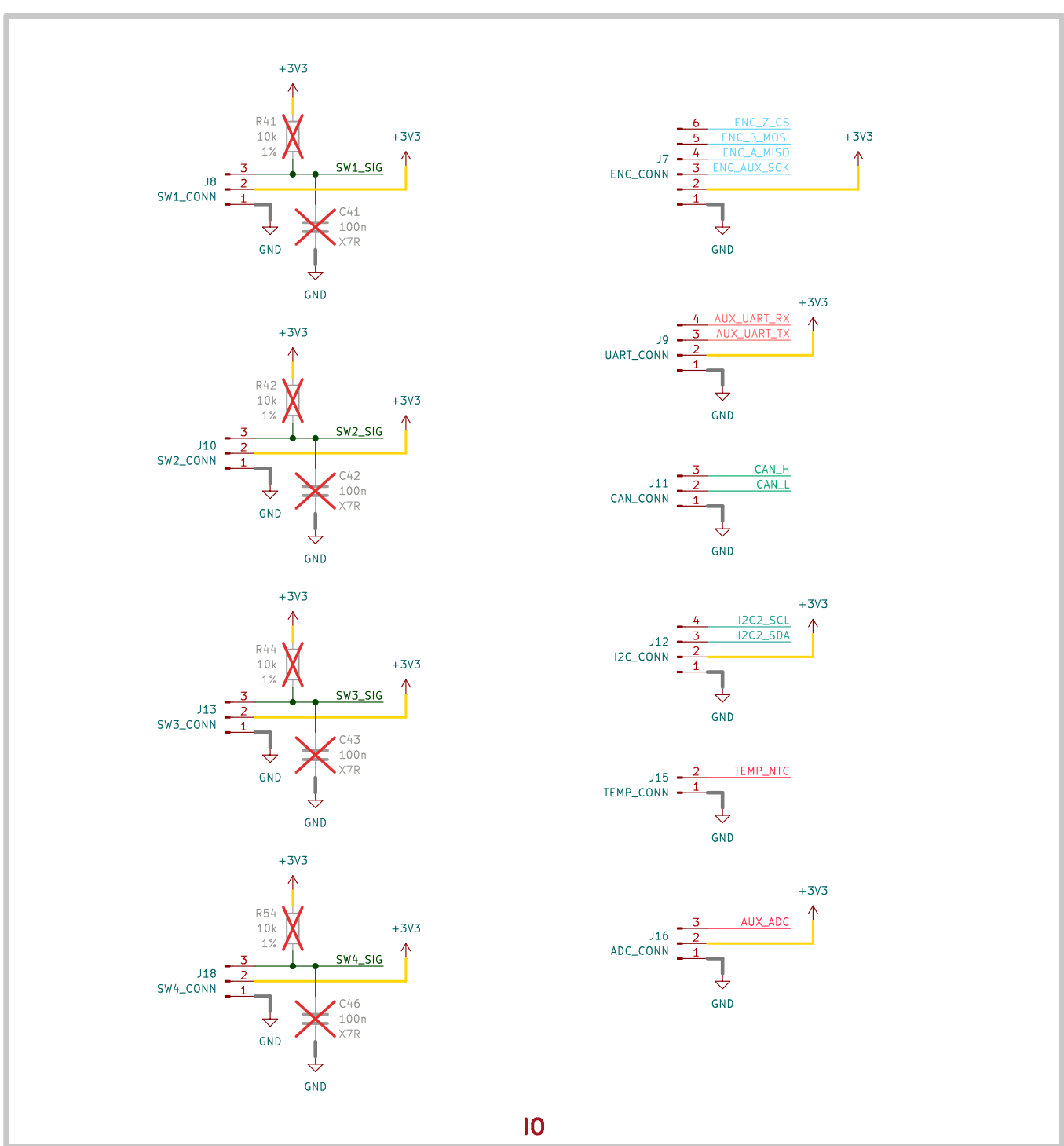
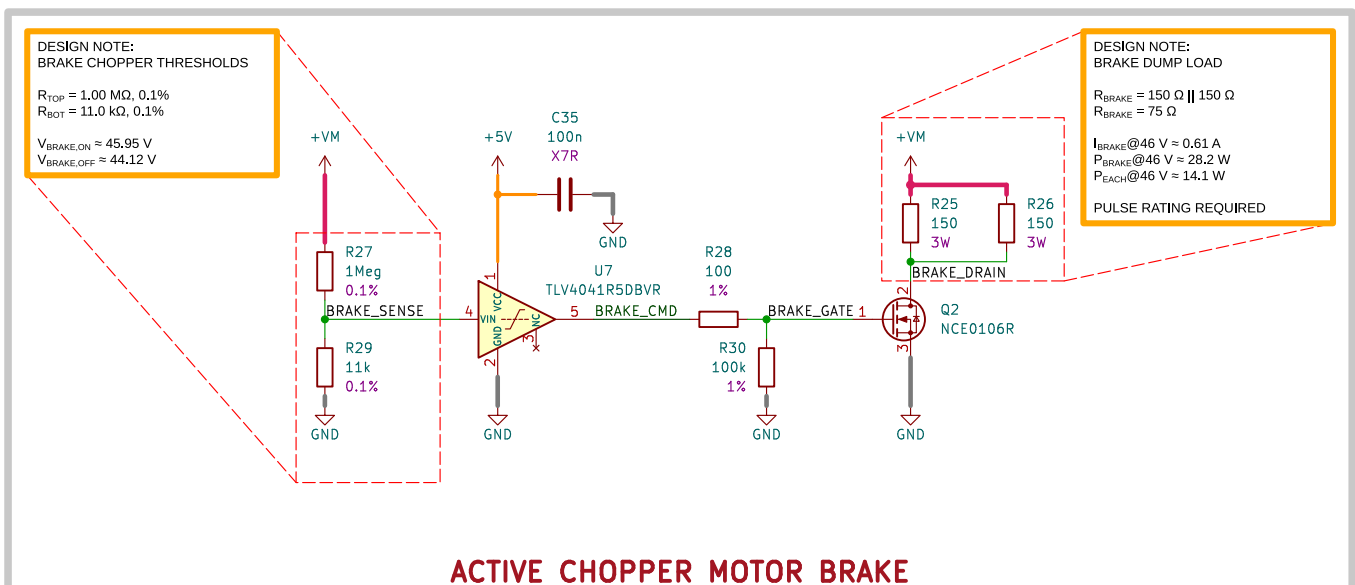
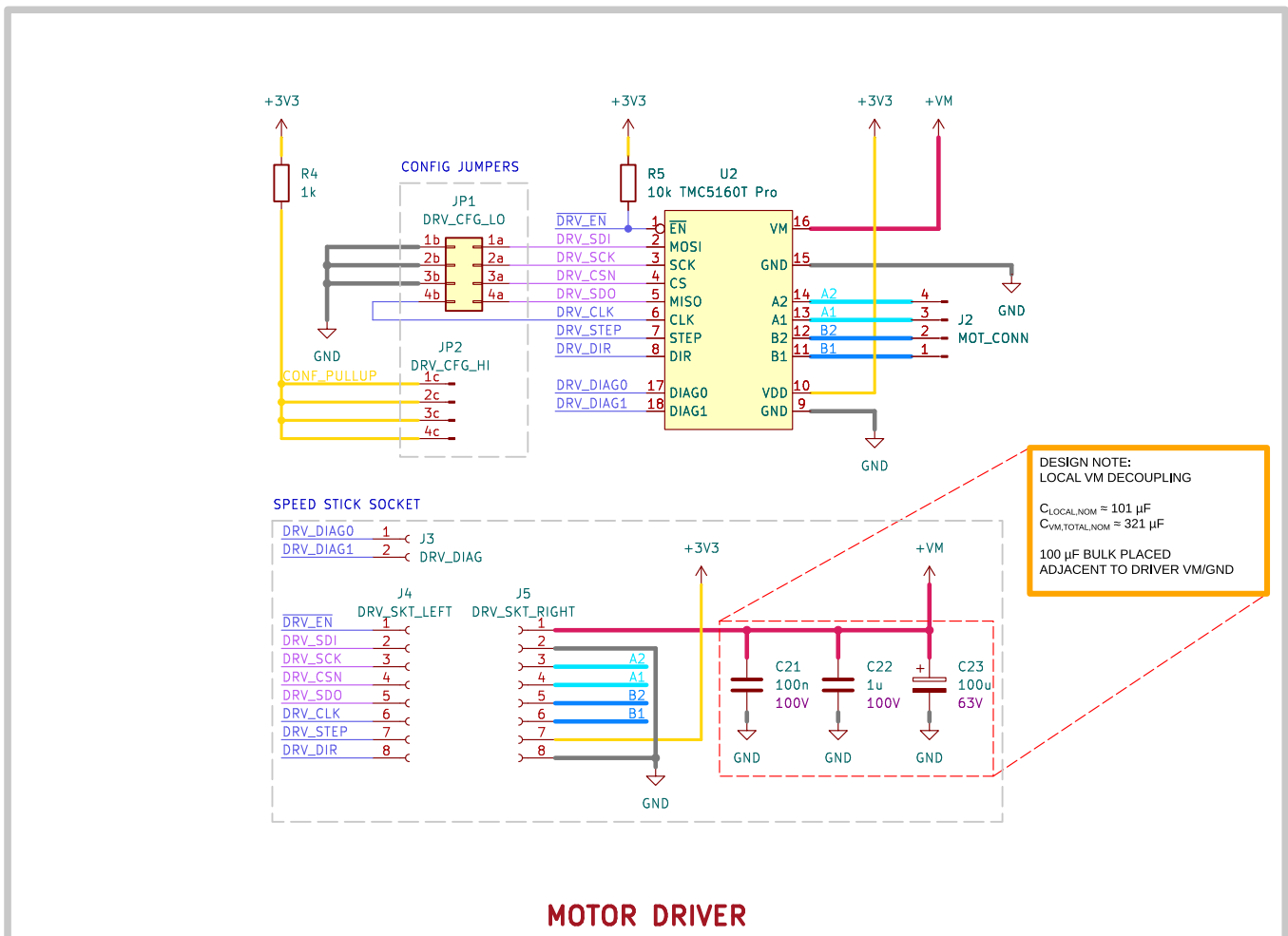
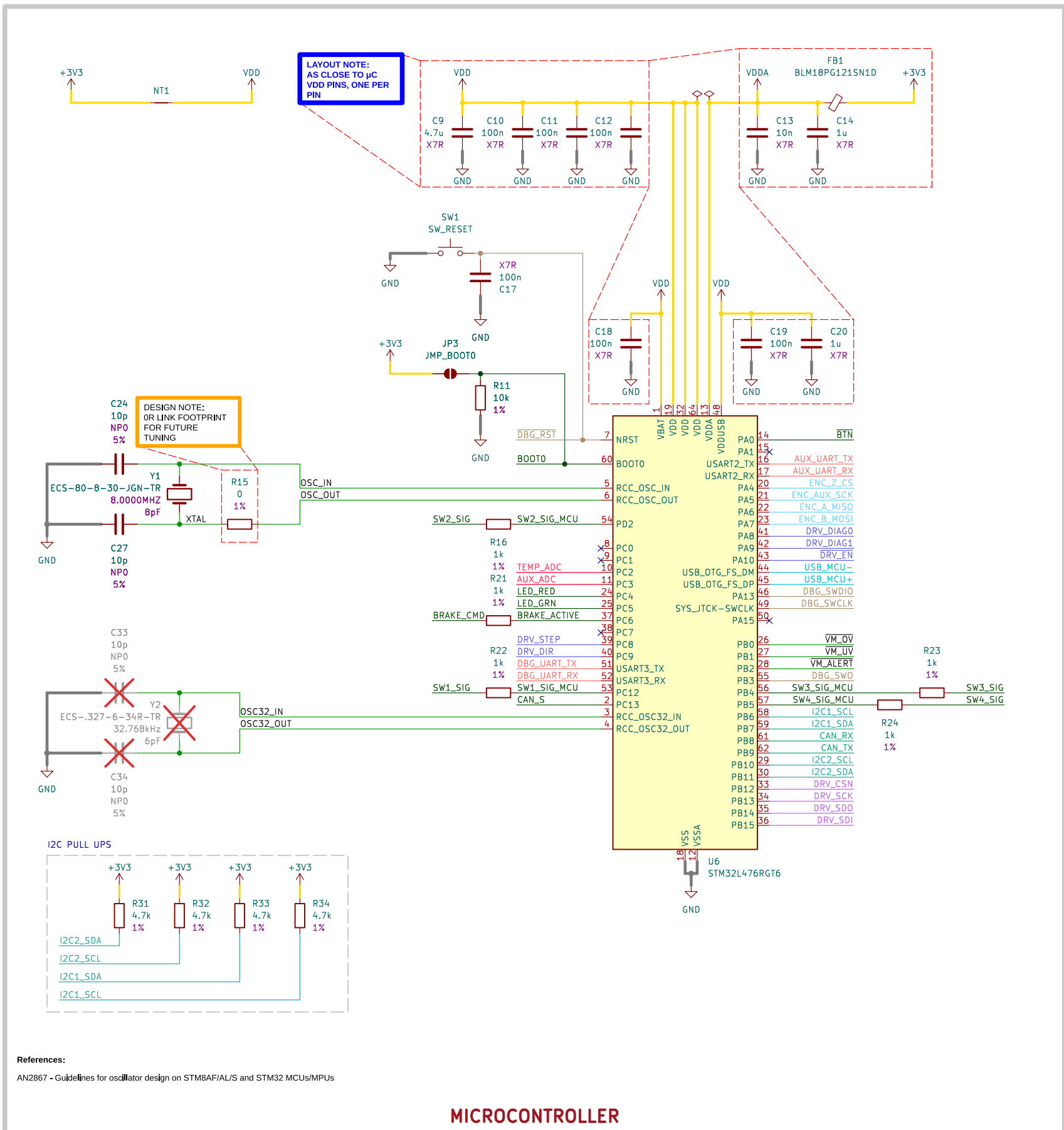
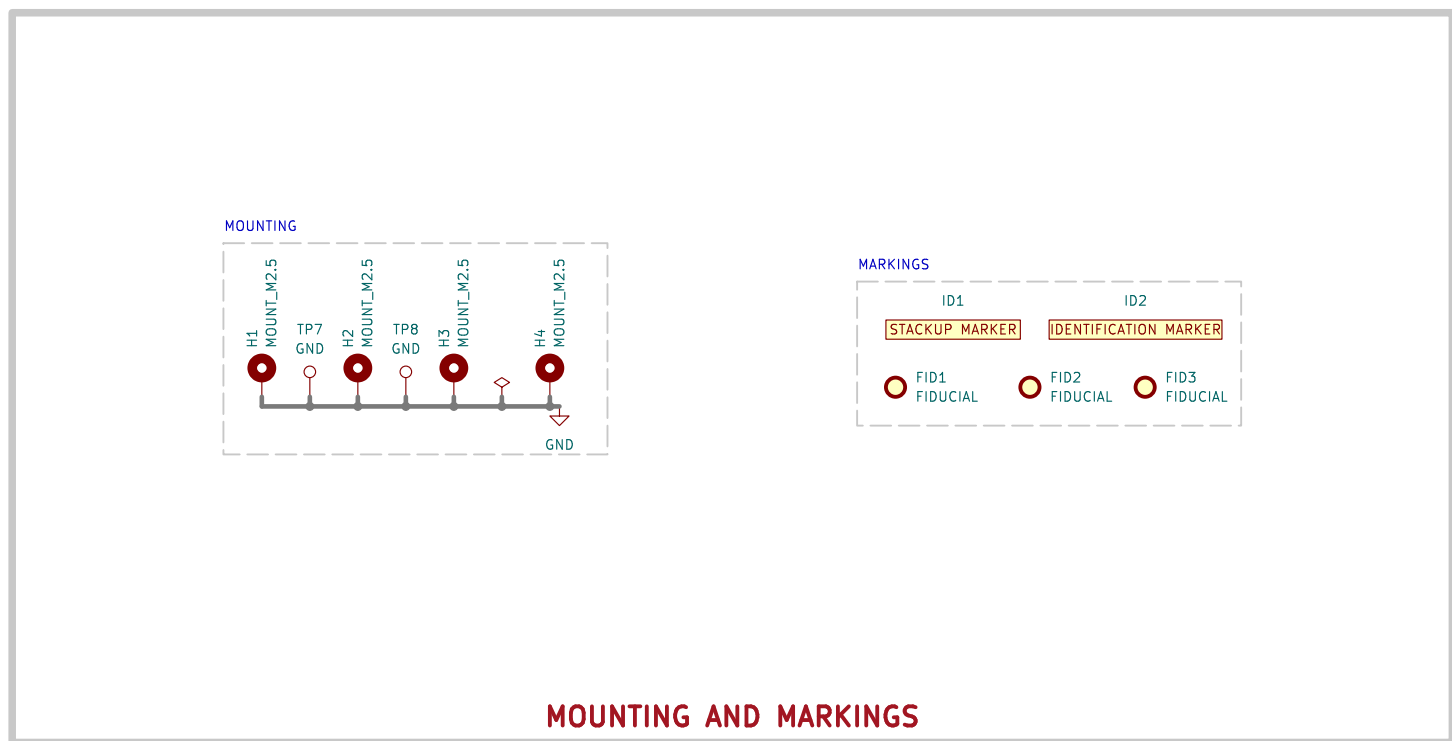
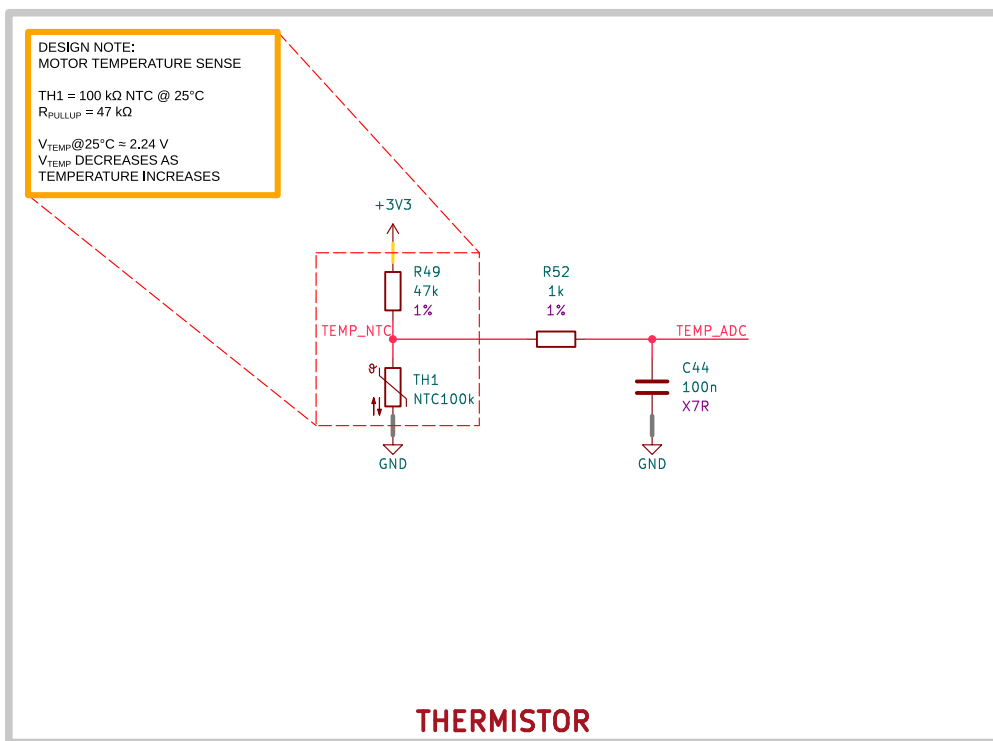
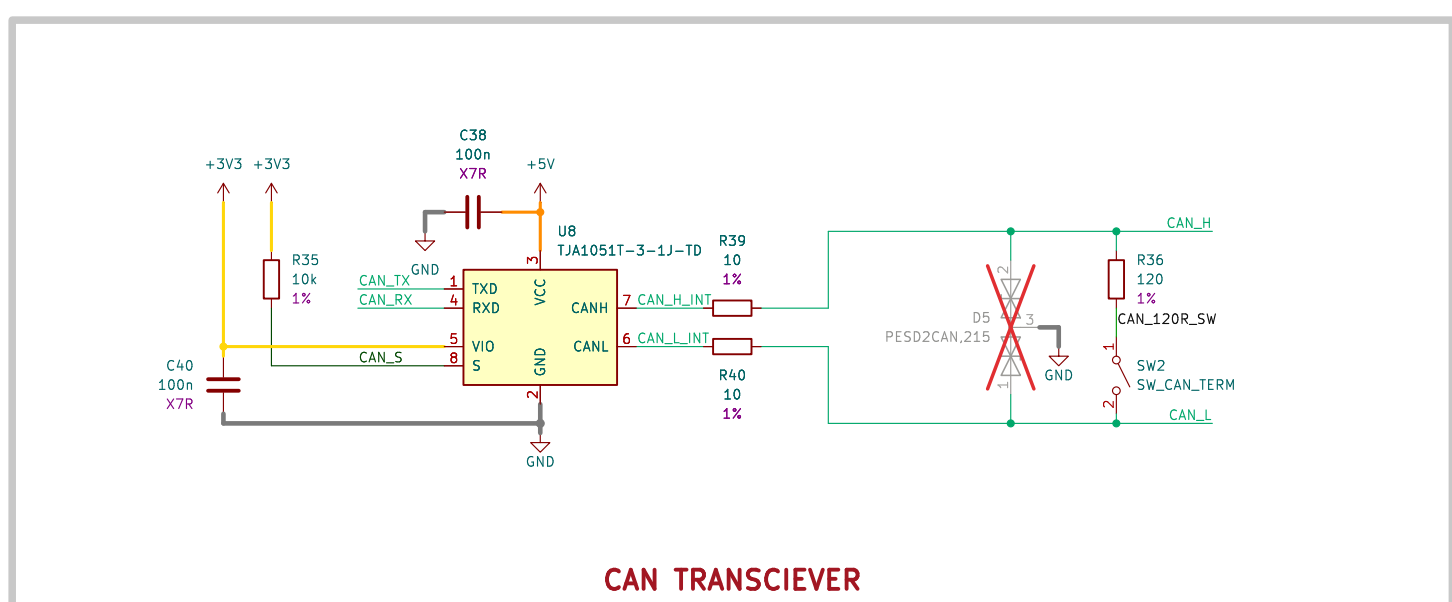
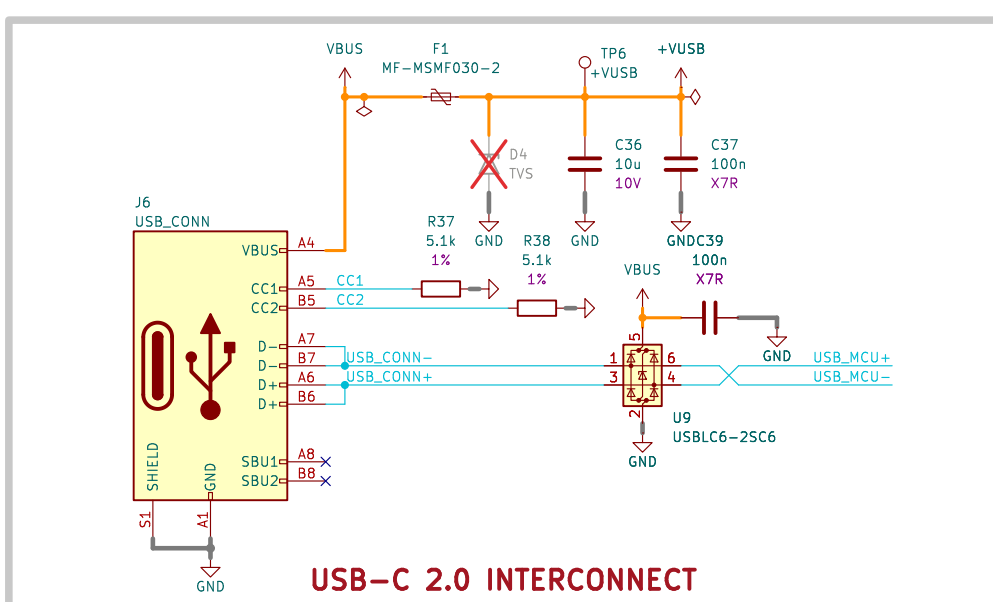
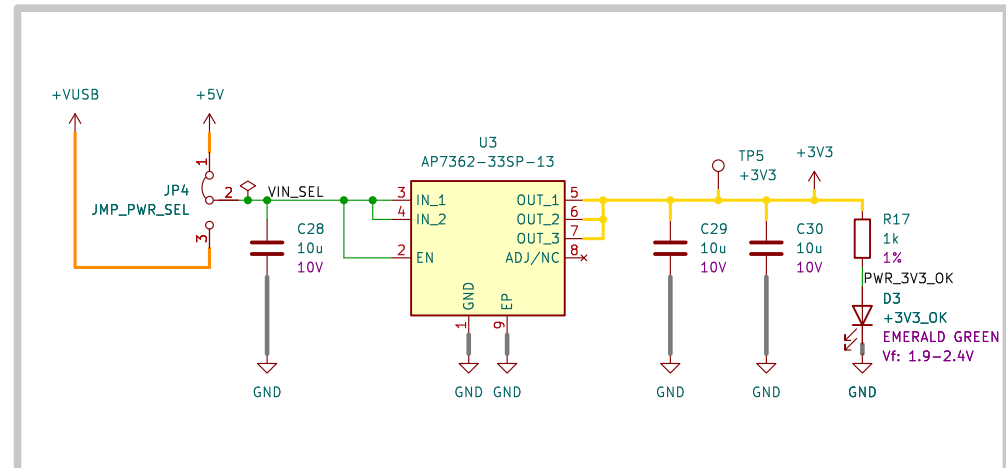
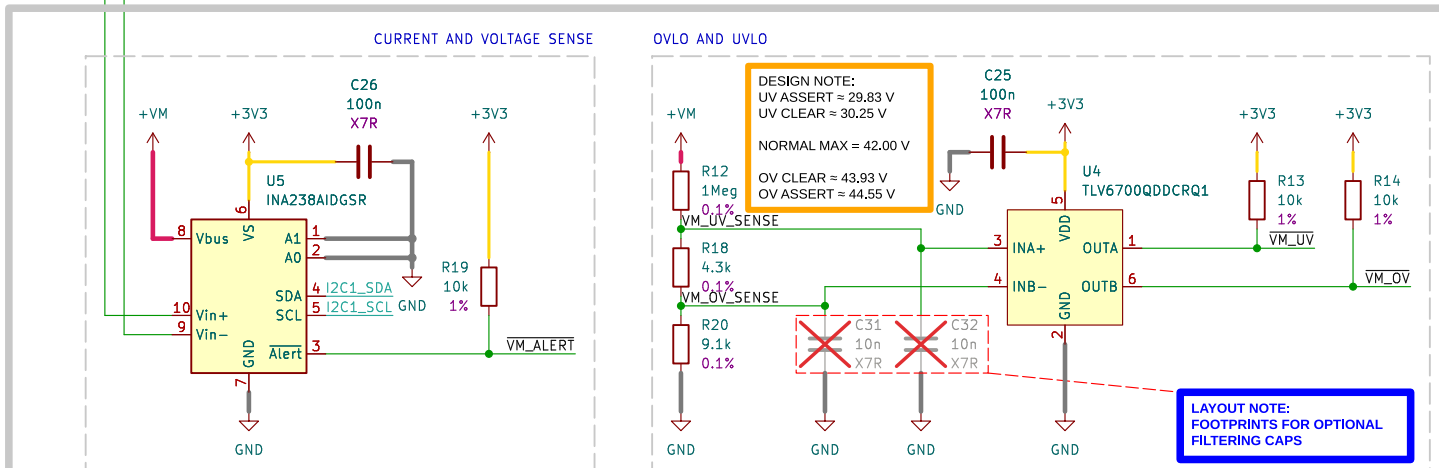
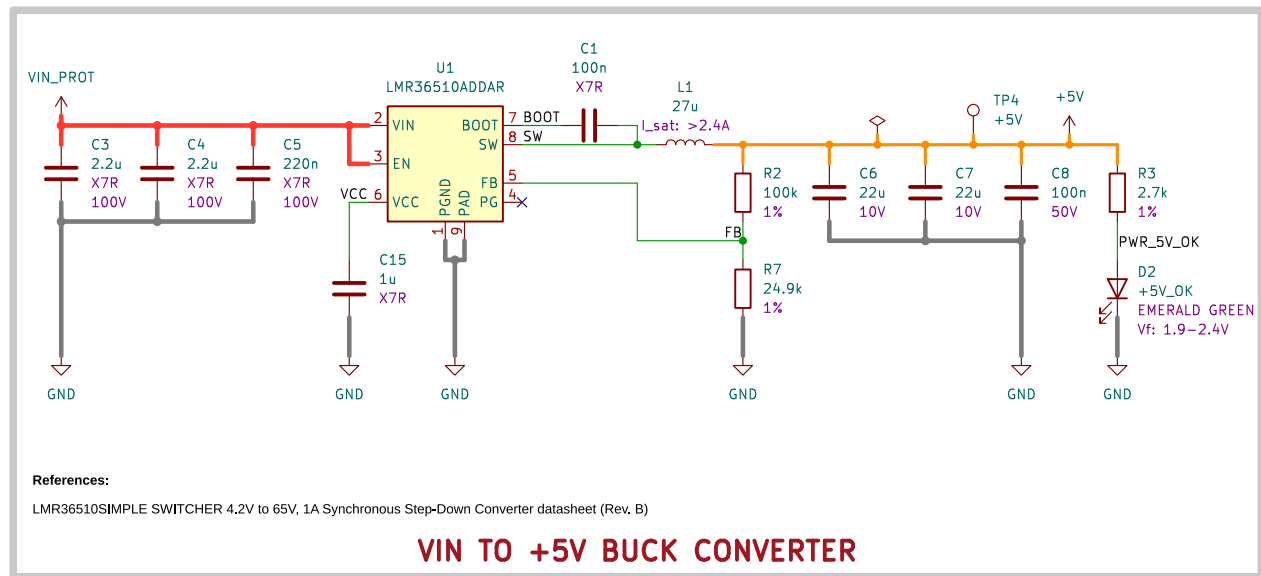
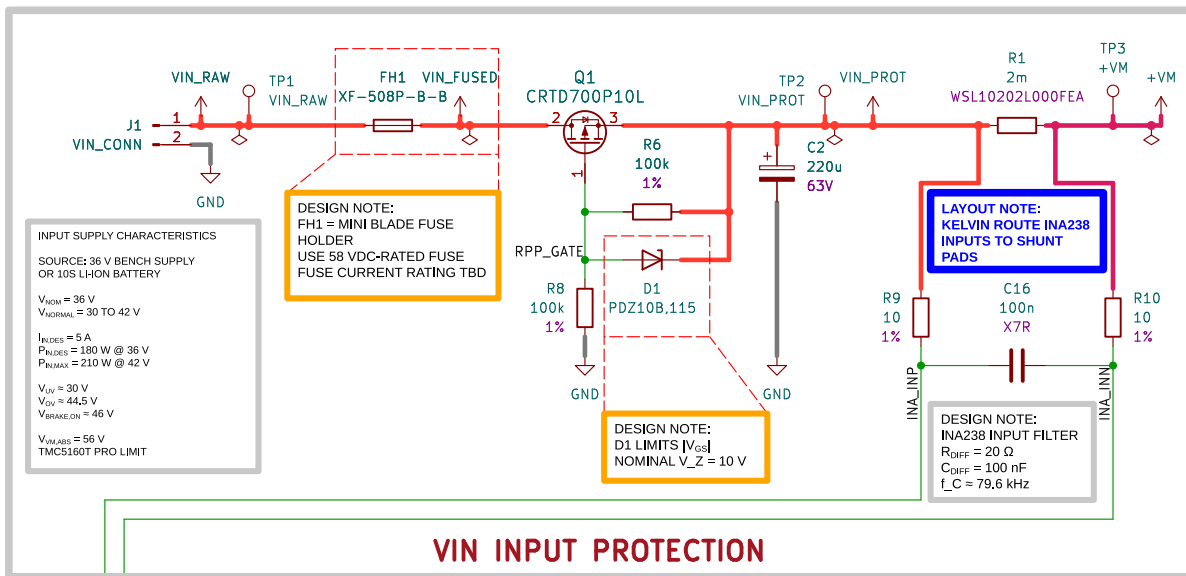
REV	DATE	DESCRIPTION	DRAWN	CHECK
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[2] Block Diagram



Target specifications:		
Input voltage:	??	– ??
Spec 2	V??	
Spec 3	??	
Spec 4	??	

					BOARDWRIGHT							
									NAME	R. HICKS		
									DATE	2026-09-27		
					SCALE NTS	SCHEMATIC		KNEE CONTROLLER				
						DRG. NO.		REV				
						4840.703		A				
REV	DATE	DESCRIPTION	DRAWN	CHECK		A3	SHEET 2 OF 10					



NAME	R. HICKS	CHECKED		APPROVED	
DATE	2026-09-27				
SCALE	NTS				
REV	A1				

BOARDWRIGHT

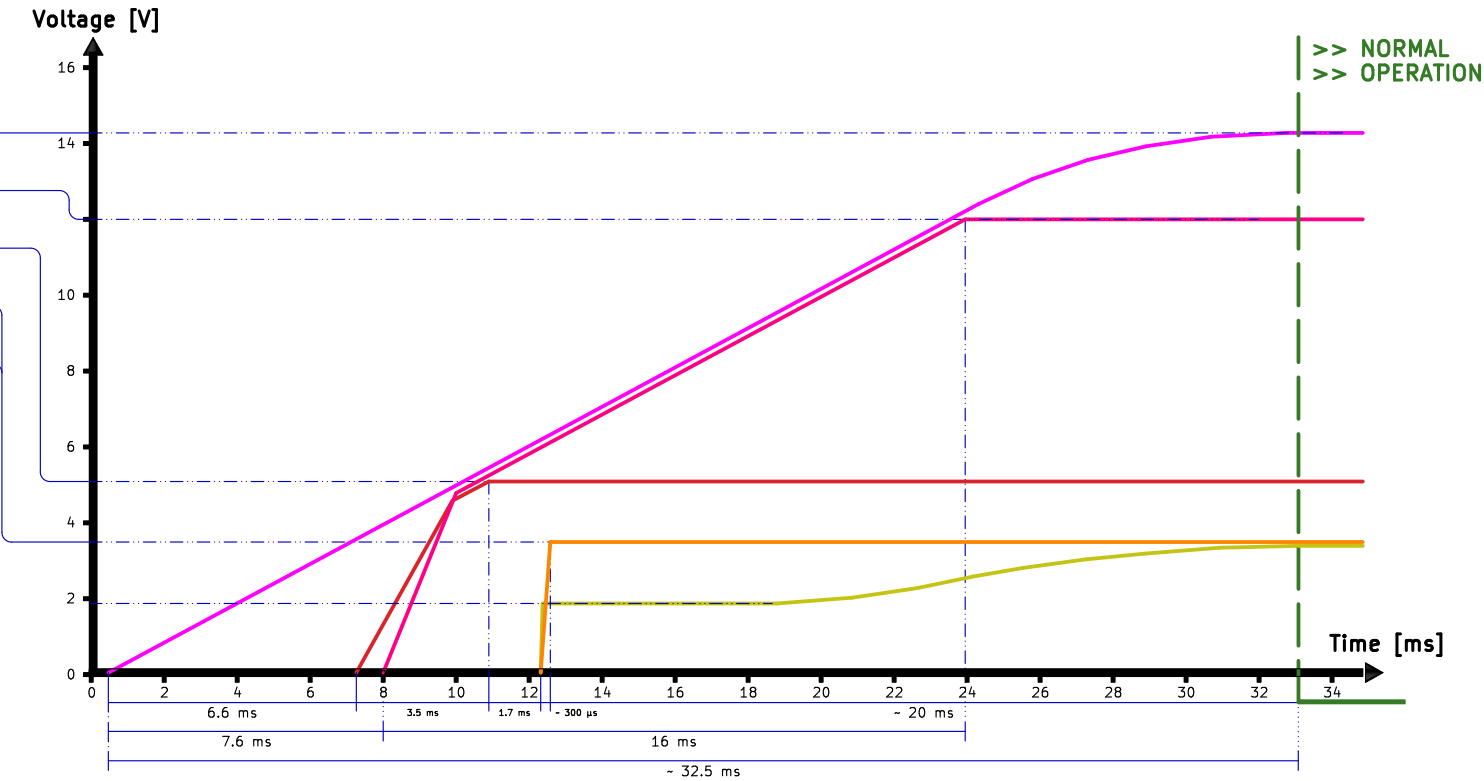
KNEE CONTROLLER

DRG. NO. 4840.703

REV A

[6] POWER – SEQUENCING

NAME	SOURCE	LEVEL
+V??	??	? – ? ± ??%
+V??	??	? – ? ± ??%
+V??	??	? – ? ± ??%
+V??	??	? – ? ± ??%
+V??	??	? – ? ± ??%

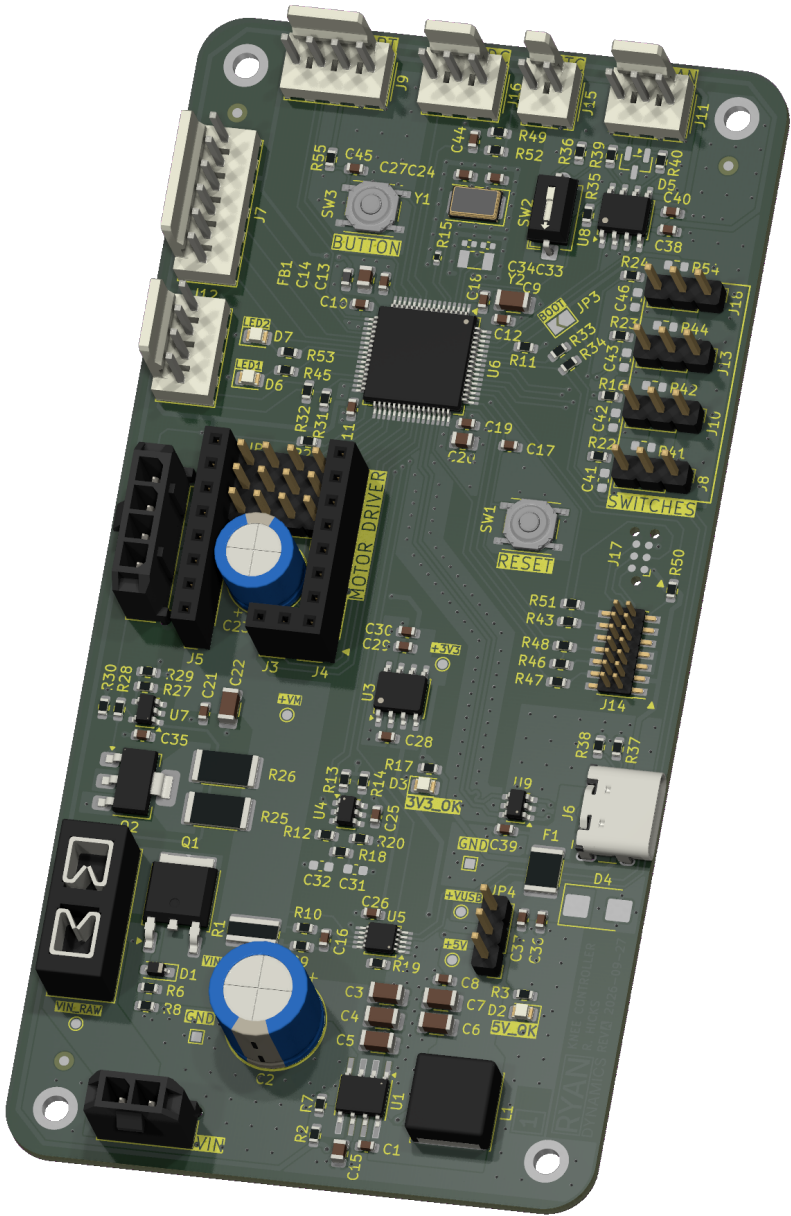


	DRAWN	CHECKED	APPROVED	BOARDWRIGHT	
NAME	R. HICKS				
DATE	2026-09-27				
SCALE NTS	SCHEMATIC			KNEE CONTROLLER	
				DRG. NO.	REV
					A
	SHEET 6 OF 10			4840.703	
	A4				

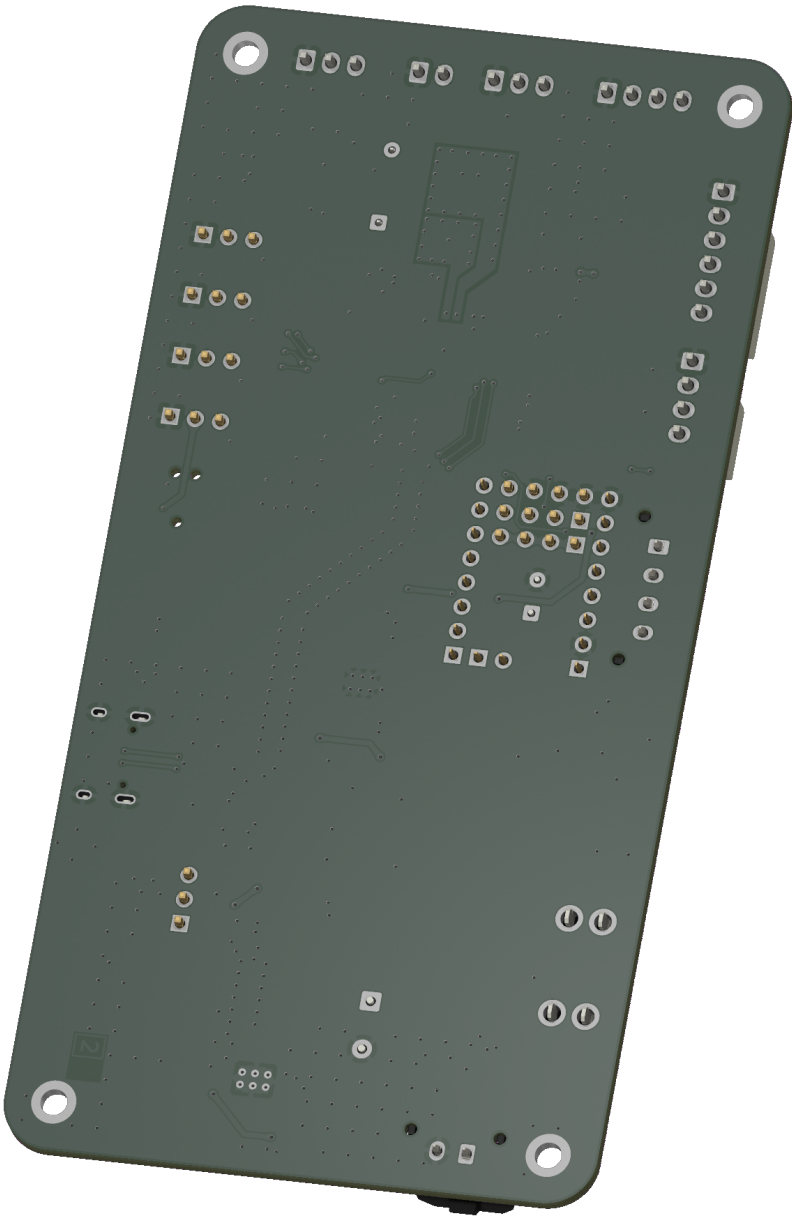
REV	DATE	DESCRIPTION	DRAWN	CHECK	

	1	2	3	4	5	6
A	[7] REVISION HISTORY					A
B						B
C						C
D					DRAWN CHECKED APPROVED	BOARDWRIGHT
					NAME R. HICKS	
					DATE 2026-09-27	
					SCHEMATIC	KNEE CONTROLLER
						DRG. NO.
						4840.703
	REV	DATE	DESCRIPTION	DRAWN	CHECK	A
	1	2	3	4	5	6

TOP VIEW



BOTTOM VIEW



REV	DATE	DESCRIPTION	DRAWN	CHECK	

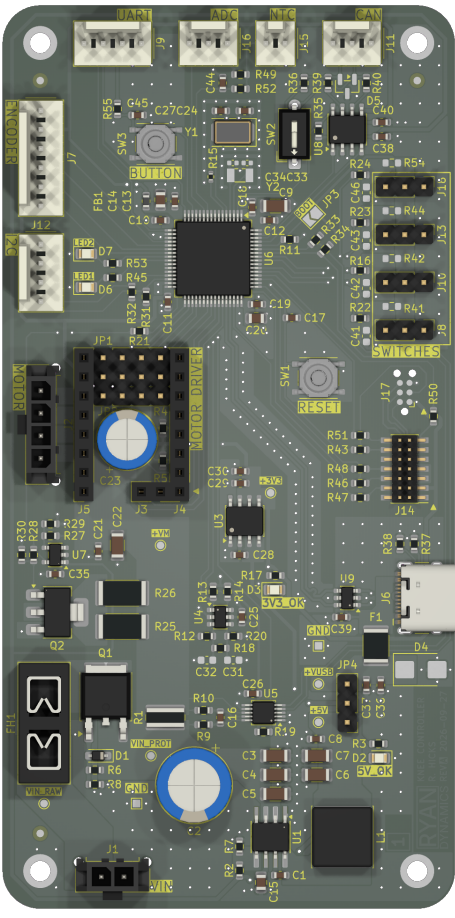
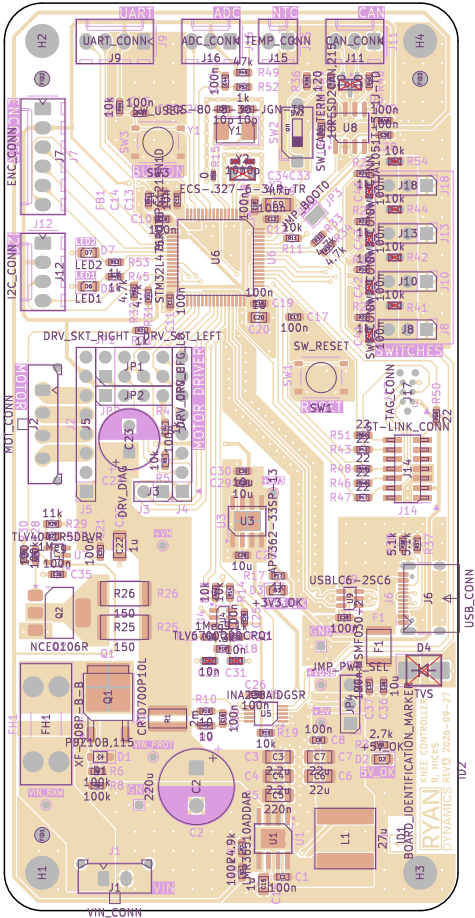
	DRAWN	CHECKED	APPROVED	BOARDWRIGHT	
NAME	R. HICKS				
DATE	2026-09-27				
SCALE NTS	ASSEMBLY OVERVIEW			KNEE CONTROLLER	
				DRG. NO.	REV
	SHEET 2 OF 10			4840.703	A

COMPONENTS

Type	Front Slide	Back Slide	Total
THT	26	0	26
SMT	142	0	142
Total	168	0	168

ASSEMBLY NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) DO NOT POPULATE PARTS ARE MARKED WITH A RED CROSS.
- 2) DO NOT POPULATE PARTS ARE NOT PRESENT IN THE BOM.
- 3) IF CONFLICTING INFORMATION IS FOUND BETWEEN THE ASSEMBLY FILE AND BOM, BOM SHOULD BE USED AS THE MAIN SOURCE.
- 4) CONFORMAL COATING IS NOT REQUIRED.
- 5) DOT IDENTIFIES PIN #1 LOCATION AND DEVICE ORIENTATION WHEN VIEWED FROM THE TOP.



	DRAWN	CHECKED	APPROVED	BOARDWRIGHT	
NAME	R. HICKS				
DATE	2026-09-27				
SCALE NTS	TOP ASSEMBLY			KNEE CONTROLLER	
	DRG. NO.			REV	
A3	SHEET 3 OF 10			4840.703 A	

		1		2		3		4		5		6		7		8					
A																				A	
B																				B	
C																				C	
D																				D	
E																				E	
F																				F	

LAYER STACKUP LEGEND

	Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste				Paste Mask	
	F.Silkscreen			Direct Printing	Legend	GBR
	F.Mask		0.02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Sig, PWR)	0.035mm (1.00oz)		Signal	GBR
	Core		1.49mm	FR4_7628	Dielectric	
	Copper	L2 (Sig, GND)	0.035mm (1.00oz)		Plane	GBR
	B.Mask		0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen			Direct Printing	Legend	GBR
	B.Paste				Paste Mask	
Total thickness: 1.6mm						
Note: external layer thicknesses are specified after plating						

IMPEDANCE TABLE

NO IMPEDANCE CONTROLLED TRACES

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

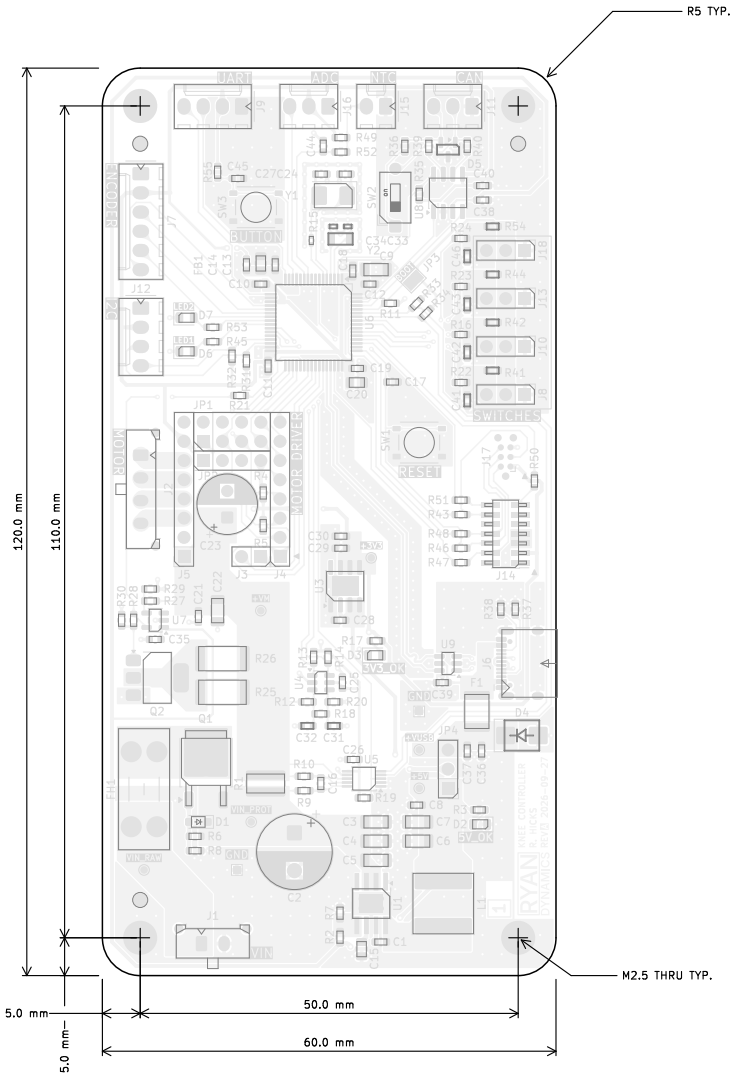
- 1) BOARD OUTLINE IS DEFINED IN THE SEPARATE GERBER FILE WITH "Edge_Cuts,GBR" SUFFIX.

DIMENSIONS OF THE CIRCUMSCRIBED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 2) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SURFACE FINISH: HAL LEAD-FREE
- 4) SOLDER MASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
- 5) SILKSCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 6) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDER MASK IS OPENED IN THE GERBER DATA.
- 7) PCB MATERIAL REQUIREMENTS:

A. CORE MATERIAL: FR-4.
- 8) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

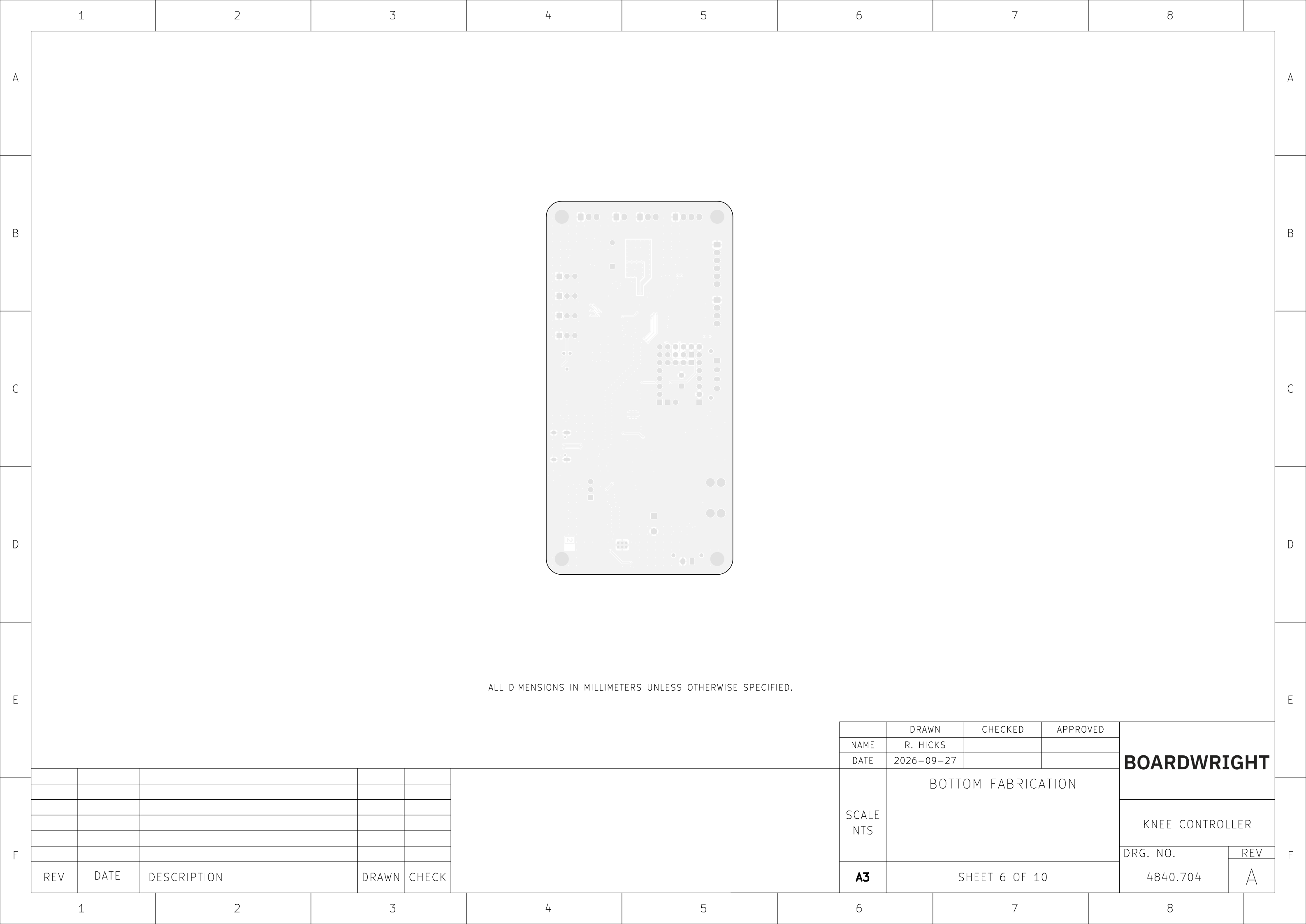
BOARD SIZE60,000 x 120,000 mm
BOARD THICKNESS1.600 mm
TRACE WIDTH0.200 mm
TRACE TO TRACE0.600 mm
MIN. HOLE (PTH)0.300 mm
MIN. HOLE (NPTH)0.650 mm
ANNULAR RING0.150 mm
COPPER TO HOLE0.254 mm
COPPER TO EDGE0.250 mm
HOLE TO HOLE0.254 mm
- 9) CONFORMAL COATING IS NOT REQUIRED.



ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

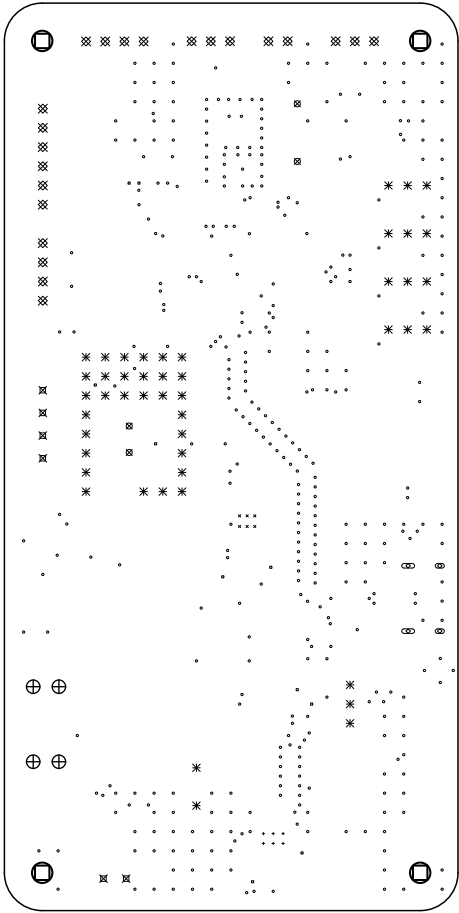
	DRAWN	CHECKED	APPROVED	BOARDWRIGHT	
NAME	R. HICKS				
DATE	2026-09-27				
SCALE NTS	TOP FABRICATION			KNEE CONTROLLER	
				DRG. NO.	REV
	SHEET 5 OF 10			4840.704	A

REV	DATE	DESCRIPTION	DRAWN	CHECK	



DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	6	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
○	406	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Via
+	6	0,33mm (12,99mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
□	10	0,40mm (15,75mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Via
◇	4	0,60mm (23,62mils)	PTH	Slot	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊗	4	0,80mm (31,50mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
✱	47	1,00mm (39,37mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⌘	6	1,02mm (40,16mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊗	22	1,19mm (46,85mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊕	4	1,78mm (70,08mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
○	4	2,70mm (106,30mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
	Total 519					

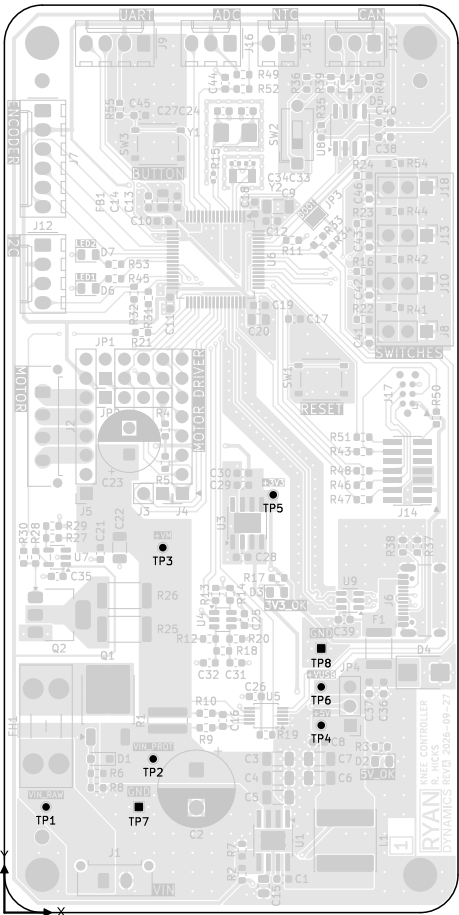


	DRAWN	CHECKED	APPROVED	BOARDWRIGHT	
NAME	R. HICKS				
DATE	2026-09-27				
SCALE NTS	L1-L2 DRILL MAP			KNEE CONTROLLER	
				DRG. NO.	REV
				4840.704	A
A3	SHEET 7 OF 10				

REV	DATE	DESCRIPTION	DRAWN	CHECK

Ref.	Net	X [mm]	Y [mm]
TP1	VIN_RAW	5.51	13.97
TP2	VIN_PROT	19.68	20.32
TP3	+VM	20.95	48.26
TP4	+5V	41.91	24.77
TP5	+3V3	35.56	55.24
TP6	+USB	41.91	29.84
TP7	GND	17.78	13.97
TP8	GND	41.91	34.92

Ref.	Net	X [mm]	Y [mm]
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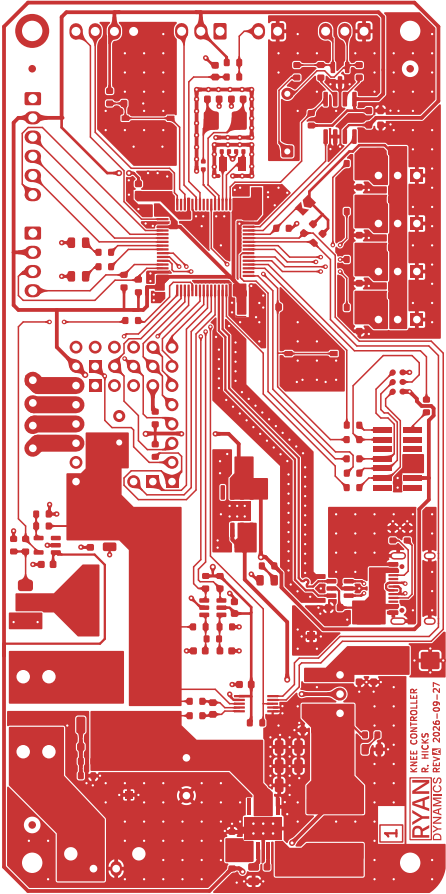


ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

	DRAWN	CHECKED	APPROVED	BOARDWRIGHT	
NAME	R. HICKS				
DATE	2026-09-27				
SCALE NTS	TOP TESTPOINTS			KNEE CONTROLLER	
				DRG. NO.	REV
	A3	SHEET 8 OF 10			4840.704

REV	DATE	DESCRIPTION	DRAWN	CHECK

	1	2	3	4	5	6	7	8		
A										A
B										B
C										C
D										D
E										E
F										F
	REV	DATE	DESCRIPTION	DRAWN	CHECK					
1	2	3	4	5	6	7	8			



	1	2	3	4	5	6	7	8		
A										A
B										B
C										C
D										D
E										E
F										F
	REV	DATE	DESCRIPTION	DRAWN	CHECK					
1		2	3	4	5	6	7	8		